From Analyst Forecasts to the Q-Theory of Corporate Investments: A Novel Methodology for Directly Estimating WACC

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Current Draft: January 8, 2025

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Abstract

This paper introduces a novel methodology for estimating the *Weighted Average Cost of Capital* (WACC) from discounted cash flow models based on analyst forecasts of operating profits. Unlike traditional methods, our approach incorporates forward-looking expectations and *directly* estimates WACC without the need to individually estimate costs of equity and debt. Cross-sectionally, our methodology assigns higher costs of capital to riskier industries, and our estimated WACC is positively related to volatility, market, SMB, and HML betas, and negatively related to market cap, leverage, debt maturity, and tax rates. Consistent with Q-theory, cross-sectional regressions reveal a significant negative relationship between WACC and future investments. We further find that high-investment firms underperform more significantly when their cost of capital is higher, rather than lower, providing new insights into investment behavior and challenging traditional risk-based explanations. In the time series, the aggregate WACC varies considerably across economic cycles and is negatively related to future investment, though this relationship weakens in the presence of the Term Spread, which itself is negatively related to future investment. Our intuitive methodology provides corporate managers a straightforward means of estimating the cost of capital for their investment decisions, while offering researchers a more reliable measure of firm-wide cost of capital.

1. Introduction

The theoretical and empirical relationship between firm-wide cost of capital, marginal Q and corporate investment decisions has long been a topic of interest in financial economics. However, the results from the empirical literature, particularly in studying aggregate investment, have been disappointing. A significant challenge lies in the lack of reliable measures of cost of capital. While data on corporate investment is readily available from standard macroeconomic and firm-specific databases, reliable estimates of cost of capital remain elusive. Estimating the cost of capital using historical realized returns has proven problematic in empirical asset pricing as standard asset pricing models that rely on historical data to estimate cost of capital often suffer from significant measurement errors.[2] Implied cost of equity capital (ICC) is a relatively new approach for estimating the cost of equity capital directly from discounted cash flow models, using analyst forecasts of earnings.[3] Cost of capital estimates derived from these methodologies have been used to study a variety of issues including cross-sectional asset pricing, time-series return predictability, intertemporal risk-return tradeoff and the default risk premium.[4]

In this paper, we introduce an implied cost of capital methodology that relies on forward looking analyst expectations of operating profits to directly estimate the weighted average cost of capital (WACC).[5] This contrasts with the traditional textbook approach, which requires separate estimates of cost of equity, cost of debt, tax rates and leverage ratios.[6] We implement this methodology using a simple rule: *leave no information behind*. In practice, this means we use what is available among Earnings Per Share (EPS), Earnings Before Interest and Taxes (EBIT) and Earnings Before Interest, Taxes, and Depreciation and Amortization (EBITDA) forecasts to derive forecasts of operating profits and growth rates (more on this in the methodology section).

[2] See Fama and French (1997) and Elton (1999).
[3] Claus and Thomas (2001), Fama and French (2002), Gebhardt, Lee and Swaminathan (2001).
[4] See Gebhardt, Lee and Swaminathan (2001), Fama and French (2002), Pastor, Sinha, and Swaminathan (2008), Lee, Ng, and Swaminathan (2009), Li, Ng, and Swaminathan (2012), Chava and Purnanandam (2010). See Lee, So, and Wang (2020) for an evaluation of alternative measures of expected returns.
[5] The weighted average cost of capital we estimate is nominal. However, we compute a real weighted average cost of capital using the Fisher model, which is the measure applied in all our empirical tests. Henceforth, when we refer to the weighted average cost of capital, we are referring to the real measure.
[6] See Jagannathan, Liberti, Liu, and Meier (2017) for an excellent summary of the state-of the-art in estimating WACC.

Our estimates of WACC exhibit intuitive cross-sectional and time-series properties. In the cross-section, for instance, our methodology assigns higher costs of capital to sectors considered riskier, such as Materials (9.69%) and Information Technology (8.62%), and lower costs of capital to less risky sectors like Utilities (4.42%) and Consumer Staples (6.23%). In the time-series, the aggregate (value-weighted) WACC seems to vary over time in an expected manner: it peaked in early 1990s, declined in the late 1990s, peaked again during the 2009-2011 time period, steadily declined until the pandemic in early 2020, when it briefly surged before declining sharply again (See Figure 1a). The aggregate WACC reached its highest value following the global financial crisis, standing at 10.51% in June 2010, and fell to its lowest value of 2.26% in March 2022.

In cross-sectional regressions, we find that WACC is positively related to return volatility, market, SMB and HML betas, and negatively related to market cap, tax rates, financial leverage and debt maturity. Thus, riskier firms face higher costs of capital and larger firms face lower costs of capital. Notably investment and profitability betas are not significantly related to WACC suggesting these risks are not important determinants of firm-wide cost of capital. The negative relationships between leverage and cost of capital as well as tax rate and cost of capital suggest that corporate interest tax shields lower cost of capital and add to value. Additionally, the (weak) negative relationship between WACC and debt maturity is consistent with the rollover risk hypothesis.[7] In terms of economic significance, volatility and tax rate command the largest cost of capital premiums, with magnitudes ranging from 3% to 4% in absolute value. Market, SMB, HML beta garner premiums in the range 0.6% to 1.7%.

Frank and Shen (2015) develop a Q-theory model that predicts a negative relationship between the weighted average cost of capital (WACC) and corporate investment, after controlling for profitability. Using the implied cost of equity as an input to the textbook WACC formula, they empirically confirm the negative relationship.[8] Our paper also finds a robust negative relationship between our direct measure of the implied weighted average cost of capital and future corporate

[7] See He and Xiong (2012).
[8] Interestingly, they also report a positive relationship between WACC and investment when the cost of equity is estimated using traditional CAPM based on historical returns. They argue that the implied cost of equity is likely a better measure of time-varying expected return.

investment after controlling for cash flows, as predicted by the Q-theory. Contemporaneous cash flows are also a significant determinant of corporate investment.

Building on the cross-sectional findings, we revisit the commonly observed underperformance of high-investment firms by conditioning this phenomenon on firms' cost of capital. While high-investment firms typically exhibit lower returns, we find that this relationship changes notably when the cost of capital is taken into account. High WACC, high-investment firms significantly underperform relative to their low-investment counterparts, suggesting that overinvestment, despite high hurdle rates, and possibly driven by over-optimism about future profitability, may contribute to this underperformance. Put another way, given their high cost of capital, they may not be profitable enough. In contrast, Low WACC-High Investment firms outperform in terms of risk-adjusted returns. These results appear inconsistent with traditional risk-based explanations. These findings underscore that the underperformance of high investment firms requires a more nuanced interpretation based on the interplay between investment behavior and cost of capital.

In their seminal paper, Abel and Blanchard (1986) investigate the relationship between aggregate corporate investment and marginal Q for the U.S. manufacturing sector, finding that the two are significantly positively related. They further find that fluctuations in marginal Q are driven more by variations in the cost of capital than by changes in marginal profits. When analyzing the two components separately in the investment regression, they find that the marginal profit component has a larger and more significant effect on investment than the cost of capital component. However, their analysis is limited to an indirect examination of the cost of capital's effect on investment through marginal Q, without directly estimating the cost of capital itself. In contrast, our study directly examines the relationship between cost of capital (WACC) and aggregate investment.

Kothari, Lewellen, and Warner (2016) find that aggregate corporate investment is largely driven by past profits and stock returns, with little connection to interest rates, the term spread, or the default spread. We confirm the strong positive relationship between profitability and future investment but also uncover a significant negative relationship between WACC and future investment. This finding addresses a persistent challenge within empirical tests of Q-theory—namely, identifying the economic determinants of cyclical variability in investment. Our results

suggest that corporate business investment is indeed sensitive to fluctuations in the aggregate cost of capital, a crucial link that has remained largely untested in the literature due to the absence of reliable measures of WACC. The inclusion of the term spread in the aggregate investment regression, however, weakens the significance of WACC, while the term spread itself is negatively related to future investment. The time-series correlation between WACC and the term spread is 61% (see Figure 3) suggesting that interest rate dynamics play a more substantial role in shaping future investment decisions than previously recognized.

In summary, our paper makes the following contributions:

- First, we introduce a novel methodology to directly estimate WACC from the free cash flow to the firm (FCFF) valuation model. This approach relies on forecasts of operating profits and reinvestment rates to estimate FCFF. This direct approach circumvents the need to separately estimate cost of equity and cost of debt, as in the traditional textbook methods.
- Second, industry costs of capital aggregated from individual firm estimates, demonstrate that riskier industries have higher WACC, while less risky industries have lower WACC. Cross-sectionally, firms with higher market, SMB, and HML betas, higher total and idiosyncratic volatility, lower financial leverage, lower tax rates and higher proportion of short-term debt relative to long-term debt have higher WACC. These findings indicate that riskier firms command higher costs of capital, while interest tax shields contribute to reducing the cost of capital.
- Third, cross-sectional regressions reveal a strong negative relationship between individual firm WACC and future corporate investment, consistent with Q-theory.
- Fourth, aggregate WACC is significantly negatively related to future aggregate investment, consistent with Q-theory. However, this relationship weakens in the presence of the term spread, which is also negatively related to future investment.

The rest of the paper proceeds as follows. Section 2 describes the empirical methodology. Section 3 describes the data used in the paper. Section 4 provides descriptive statistics on the cost of capital estimates and evaluates their cross-sectional determinants. Section 5 reports results on the cross-sectional relationship between investment and cost of capital. Section 6

reports average monthly returns of univariate and bivariate portfolios based on investment and WACC. Section 7 reports aggregate time-series results. Section 8 concludes.

2. Methodology

The enterprise value of a company is the present value of its free cash flows to the firm (FCFF) discounted at its weighted average cost of capital (WACC):

$$V_t = \sum_{k=1}^{\infty} \frac{FCFF_{t+k}}{(1 + WACC)^k} \tag{1}$$

Free cash flows to the firm (FCFF) are defined as:

$$FCFF_t = NOPAT_t + Dep_t - Capex_t - \Delta WC_t \tag{2}$$

where:

$NOPAT_t$	=	$EBIT_t\ (1\text{-}T)$
$EBIT_t$	=	Earnings Before Interest and Taxes
T	=	Tax rate
Dep_t	=	Depreciation & Amortization
$Capex_t$	=	Capital Expenditures
ΔWC_t	=	Change in operating working capital (inventories, receivables, payables)

We can rewrite equation (2) as follows:

$$FCFF_t = NOPAT_t - I_t \tag{3}$$

where $I_t = Capex_t + \Delta WC_t - Dep_t$ is the net new investments (net of depreciation which is a proxy for replacement investment). Define reinvestment rate as $b_t = I_t / NOPAT_t$ and rewrite equation (3) as:

$$FCFF_t = NOPAT_t(1 - b_t) \tag{4}$$

The infinite horizon model in equation (1) can now be implemented as the sum of the present value of free cash flows up to a finite horizon *T* and the present value of a terminal value (TV) at time *T* which, in turn, is the present value of free cash flows beyond year *T* (from *T+1* to infinity). Putting this all together, equation (1) can now be rewritten in its implementable form as follows:

$$V_t = \sum_{k=1}^{T} \frac{NOPAT_{t+k}(1 - b_{t+k})}{(1 + WACC)^k} + \frac{TV_T}{(1 + WACC)^T} \tag{5}$$

The terminal value is computed from a constant growth model which assumes that the free cash flows in the *steady state* (starting in year *T*+1) grow at a constant rate *g* forever. This steady state growth is accompanied by a constant reinvestment rate of *b* and a *marginal rate of return on new investment* of *ROI* such that $g = b \times ROI$. This relationship is referred to as the *sustainable growth rate*. Thus, $b = g/ROI$ and:

$$\begin{aligned} TV_T &= \frac{NOPAT_{T+1}(1 - b)}{WACC - g} \\ TV_T &= \frac{NOPAT_{T+1}(1 - g/ROI)}{WACC - g} \end{aligned} \tag{6}$$

If *ROI=WACC*, i.e., the marginal rate of return on new investment equals cost of capital, the constant growth model in equation (6) simplifies to a no growth perpetuity model: $TV_T = NOPAT_{T+1}/WACC$. This does not mean that there is no growth in operating profits, it simply means that any such growth is value-irrelevant, i.e., growth does not add to firm value.

The model in equation (5) requires explicit forecasts of *NOPAT* and *b* up to year *T+1* to estimate the enterprise value. To estimate *NOPAT* for year *t+1*, we rely on analyst forecast data, while the reinvestment rate b_{t+1} for year *t+1* is derived from historical data. *NOPAT* for year *t+2* is estimated

using the formula, $NOPAT_{t+2} = NOPAT_{t+1} \times (1+g_{t+2})$ where the growth rate g_{t+2} is also estimated from analyst forecast data (see Section 2.1). The steady-state growth rate g (starting in year T+2) is assumed to be the long-run nominal GDP growth rate estimated as the average of the annual nominal GDP growth rates up to the prior calendar year. The steady-state reinvestment rate b is g/ROI and assuming $ROI = WACC$ in the steady-state (competition drives economic profits to zero), $b = g/WACC$. Since we know the initial and final growth rates and reinvestment rates, we can interpolate the intermediate values. We interpolate growth rates based on an exponential rate of decline and the reinvestment rates based on a linear decline. Thus, g_{t+2} exponentially converges to g and b_1 linearly converges to b.

For $t+3 \leq t+k \leq t+T+1$:

$$\begin{aligned} g_{t+k} &= g_{t+k-1} \times exp[ln(g/g_2)/T] \\ NOPAT_{t+k} &= NOPAT_{t+k-1} \times (1 + g_{t+k}) \end{aligned} \tag{7}$$

For $t+2 \leq t+k \leq t+T$:

$$b_{t+k} = b_{t+k-1} - \frac{b_{t+1} - b}{T} \tag{8}$$

Using equations (5) to (8), we can directly estimate the *WACC* of a firm as the *internal rate of return (IRR)* that equates the present value of free cash flows from operations *(FCFF)* on the right-hand side of equation (5) to the enterprise value V_t on the left-hand side.

2.1 Analyst forecasts of EBITDA, EBIT, and EPS

Consensus analyst *EPS* forecasts are available for more years and for more firms than consensus *EBIT* and *EBITDA* forecasts. Even when all three forecasts are available, the number of analysts issuing *EPS* forecasts is generally higher than the number of analysts issuing *EBIT* or *EBITDA* forecasts: $N_{EPS} \geq N_{EBIT}$ *and* $N_{EPS} \geq N_{EBITDA}$ where N refers to the number of analysts. Therefore, to not throw away any forecast information, we develop a methodology for constructing *EBIT* and *EBITDA* forecasts from *EPS* forecasts.[9] We outline this methodology below:

[9] We use *EBITDA* forecasts to estimate the second-year growth rate as will be made clear later in this section.

$$EBITFY_{C,1} = (EPSFY_1 \times ShrsOutIBES) \times \frac{EBIT_0}{EPS_{OP,0} \times ShrsOut}$$
$$EBITFY_{C,2} = (EPSFY_2 \times ShrsOutBES) \times \frac{EBIT_0}{EPS_{OP,0} \times ShrsOut} \quad (9)$$

$EBITFY_{C,1}$ and $EBITFY_{C,2}$ are *EBIT* forecasts constructed from *EPS* forecasts, $EPSFY_1$ and $EPSFY_2$. $EBIT_0$ is the trailing four quarter *EBIT* from Compustat and $EPS_{OP,0}$ is the trailing four quarter Operating Earnings Per Share which represents earnings before special items (as identified by Compustat).[10] Where $EPS_{OP,0}$ is not available, we use trailing four quarter aggregate earnings before special items in the denominator of equation (9).[11] *ShrsOutIBES* is the shares outstanding from *IBES* and *ShrsOut* is the shares outstanding from Compustat corresponding to the operating earnings per share. Please refer to **Appendix** for a complete list of Compustat data items and their identifiers. We obtain earnings and shares outstanding from the same database to ensure consistency. The key assumption behind the constructed estimates in equation (9) is that the ratio of *EBIT/EBITDA* Profit Margin (*EBIT/Sales or EBITDA/Sales*) to Net Profit Margin (*Net Income Before Special Items/Sales*) remains constant over the next two years. This allows us to construct *EBIT* and *EBITDA* forecasts from available *EPS* forecasts without throwing away valuable information. The calculations for *EBITDA* forecasts are performed in a similar manner:

$$EBITDAFY_{C,1} = (EPSFY_1 \times ShrsOutIBES) \times \frac{EBITDA_0}{EPS_{OP,0} \times ShrsOut}$$
$$EBITDAFY_{C,2} = (EPSFY_2 \times ShrsOutIBES) \times \frac{EBITDA_0}{EPS_{OP,0} \times ShrsOut} \quad (10)$$

We now compute a weighted average of the constructed and actual forecasts to come up with the final forecasts. For $N_{EPS} \geq N_{EBIT}$ and $N_{EPS} \geq N_{EBITDA}$:

[10] Analyst forecasts are also before special items, so it makes sense to use trailing earnings before special items.
[11] We construct it as the trailing four quarter of *ibq-spiq (1-tax rate)*, where *ibq* is income before extraordinary items, and spiq is special items.

$$EBITFY_1 = EBITFY_{C,1} \times \frac{NEPS_1 - NEBIT_1}{NEPS_1} + EBITFY_{A,1} \times \frac{NEBIT_1}{NEPS_1}$$
$$EBITFY_2 = EBITFY_{C,2} \times \frac{NEPS_2 - NEBIT_2}{NEPS_2} + EBITFY_{A,2} \times \frac{NEBIT_2}{NEPS_2} \quad (11)$$

where $EBITFY_{A,1}$ and $EBITFY_{A,2}$ are the actual consensus analyst forecasts. Since, in our sample, N_{EPS} is generally higher than N_{EBIT} or N_{EBITDA}, constructed forecasts are weighted by the additional number of analysts who make *EPS* forecasts but not *EBIT* or *EBITDA* forecasts and the actual *EBIT* or *EBITDA* forecasts are weighted by the number of analysts who do make those forecasts. To avoid repetition, we leave out the corresponding equations for weighted *EBITDA* forecasts. Simply replace *EBIT* by *EBITDA* and *NEBIT* by *NEBITDA* in equation (11). We require $EBITFY_1 > 0$ and $EBITFY_2 > 0$ and $EBITDAFY_1 > 0$ and $EBITDAFY_2 > 0$. Firms for whom the number of *EBIT* or *EBITDA* forecasts are greater than the number of *EPS* forecasts we simply use the actual *EBIT* or *EBITDA* forecasts.

We construct 12-month ahead forecasts from the fiscal year forecasts by weighting by the number of months left to the fiscal year-end:

$$EBIT_1 = w \times EBITFY_1 + (1 - w) \times EBITFY2 \quad (12)$$

where *w* is the ratio of the number of months remaining until the next fiscal-year end divided by 12. Thus, if only one month remains until the fiscal year-end, the *FY1* forecast is given a weight of 1/12, while the *FY2* forecast receives a weight of 11/12.

How do constructed *EBIT* forecasts compare to actual *EBIT* forecasts for firms where both are available? To explore this, we compare the constructed $EBITFY_1$ and analysts forecasted $EBITFY_1$. The sample means for the constructed and actual forecasts are \$969.6 million and \$966.9 million, respectively, while the coefficients of variation for the two forecasts are 4.5 and 3.7. This indicates that *EBIT* forecasts derived from EPS forecasts exhibit a slightly higher variance. The Spearman rank correlation between the two forecasts across all firms and months is notably high at 92%. We have similar results for constructed $EBITFY_2$ and analysts forecasted $EBITFY_2$, with a Spearman rank correlation of 91%. Overall, the constructed forecasts align closely with actual

forecasts, suggesting that this methodology is a reasonable approach to estimating *EBIT* when direct forecasts are unavailable.

We estimate g_{t+2} as the growth rate implicit in $EBITDAFY_1$ and $EBITDAFY_2$. Thus, $g_{t+2} = EBITDAFY_2/EBITDAFY_1 - 1$. We use the implied growth rate in *EBITDA* forecasts over the implied growth rate in *EBIT* forecasts because growth rates in *EBITDA* are more persistent than the growth rates in *EBIT*, which in turn, are more persistent than growth rates in *Net Income*.[12] We estimate the next 12-month *EBIT* forecast as $EBIT_2 = EBIT_1(1+g_2)$. *We winsorize the values of* g_2 *to be between 0.01 and 0.75*.[13]

Reinvestment rate in year 1, b_1 is estimated as follows:

$$b_1 = \frac{CapEx + CashAcq + (\Delta Rec + \Delta Inv - \Delta AcctPay) - Dep\ \&\ Amort}{NOPAT_0} \qquad (13)$$

where $NOPAT_0 = EBIT \times (1 - Tax\ Rate)$. As mentioned earlier, all accounting items are based on trailing four quarter cash flow statements and are described in detail in **Appendix**. Briefly here, *CapEx* refers to capital expenditures, *CashAcq* refers to cash acquisitions, *ΔRec*, *ΔInv*, and *ΔAcctPay* refer to change in receivables, change in inventory, and change in accounts payable respectively, and *Dep & Amort* refers to depreciation and amortization expenses. Equation (13) computes the ratio of *net new investments* to after-tax operating profits which measures the fraction of operating profits reinvested by the firm. In computing b_1, to avoid dividing by a low base, we assign the median reinvestment rate of the corresponding industry (based on 4-digit GICS codes) to firms whose $NOPAT_0$ */Enterprise Value* is < 0.05%.

We also winsorize the values of b_1 *to be between 0.1 and 1.5*. A low value implies very little net new investment and a high value greater than 1 reflects a growing company reinvesting more than it earns through external financing. The tax rate is computed as the effective tax rate from the most recent fiscal year (Total income taxes/Pretax income). We set the tax rate to be between 0.1 and

[12] Generally, further up a profit measure is in the income statement the more persistent is its growth rate, see Chan, Karceski, and Lakonishok (2003)
[13] Results based on WACC estimates using EBIT based growth rates are not appreciably different.

0.6. As a robustness check, we also use the statutory tax rate of 26% (21% federal corporate rate +5% average state tax rate).

The enterprise value V is computed as follows:

$$V = MV + Debt + Minority\ Interest + Preferred\ Stock - Cash\ and\ Short\ Term\ Investments \tag{14}$$

MV is the market value of equity as of the estimation date (CRSP closing price × CRSP shares outstanding), *Debt* represents the book value of all short-term and long-term interest-bearing debt from the most recent quarter, *Minority Interest*, *Preferred Stock*, and *Cash and Short-Term Investments* all represent book values from the most recent quarter. The most recent quarter is defined as the quarter ending at least *three* months prior to the estimation date. In essence, we assume that it will take three months after the quarter-end for that quarter's accounting data to become public. Thus, for the implied *WACC* estimated on June 30, market value of equity is computed as of June 30, all income statement and cash flow statement items are computed for a trailing four quarter period ending March 31, and the balance sheet items are for the quarter ending March 31. The estimations on July 31 and August 31 will continue to use the accounting data as of March 31. The estimation on September 30 will use data as of the quarter ending June 30, and so on. We assume T=15 years and estimate the implied *WACC* for every U.S. publicly traded company with available data using the model in equation (5). The sample period is January 1989 to December 2022.

3. Data

We construct our dataset by combining CRSP, Compustat and IBES. The month-end market capitalization data is obtained from CRSP, calculated by multiplying the closing price (PRC) by shares outstanding (SHROUT). Our sample includes only common stocks listed on NYSE/AMEX/NASDAQ with share codes 10 and 11. Additionally, we exclude financial firms with SIC codes between 6000 and 6999. The sample starts in January 1989 because of the non-

availability of statement of cash flow items used in the calculation of the reinvestment rate prior to July 1988.[14]

Analyst forecasts are obtained from IBES. The aggregate earnings/net income forecast is computed by multiplying the median analyst's earnings per share (EPS) forecast (MEDEST) by the total shares outstanding (SHOUT) from IBES. Analyst forecasts of (aggregate) EBIT and EBITDA are directly sourced from the IBES summary file.

Table 1 provides annual summary statistics for the sample of firms used in this paper as of the end of June from 1989 to 2022. Listed Firms is the number of publicly traded firms with common stock listed in NYSE/AMEX/Nasdaq. IBES EPS Forecasts, IBES EBIT Forecasts and IBES EBITDA Forecasts denote the number of firms with corresponding consensus forecasts available in the IBES database. EPS Forecasts, EBIT Forecasts and EBITDA Forecasts represent the number of firms with positive FY1 and FY2 forecasts for which WACC estimates can be calculated, excluding the top and bottom 0.5% of those estimates; this set of firms constitutes the sample used in this paper. EPS Analysts, EBIT Analysts and EBITDA Analysts represent the median number of analysts making the respective forecasts within the sample.

Not surprisingly, consensus EBIT and EBITDA forecasts are available only from 2000/2001. Also, the number of firms with available EBIT and EBITDA forecasts are generally fewer than those with available EPS forecasts. The data also shows fewer analysts provide EBIT and EBITDA forecasts than do EPS forecasts. This supports our strategy of constructing EBIT and EBITDA forecasts from EPS forecasts in order not to lose any information.

For accounting data, we rely on COMPUSTAT, utilizing the items listed in **Appendix**. To avoid look-ahead bias and ensure that all financial accounting information is publicly available when utilized, we assume that all accounting information becomes public only three months after the quarter's end. Thus, for a December 31 WACC estimation, we utilize only the accounting data for the quarter ending September 30. Data on nominal GDP growth rates are obtained from the Bureau

[14] SFAS 95 related to the modern statement of cash flows was issued by the FASB in November 1987 and went into effect in July 1988.

of Economic Analysis. Our GDP data begins in 1930. For each year, we compute the steady-state GDP growth rate as the historical average of nominal GDP growth rates using annual data up to the prior calendar year. All interest rates used to compute term spread and default spread, BAA yield, AA yield, and 3-month T-bill yield, are obtained from FRED (Federal Reserve Economic Data from the St. Louis Federal Reserve Bank).

We compute real WACC using the Fisher equation: *Real WACC = (Nominal WACC-inf)/(1+inf)*, where *inf* is the average of past 12-month annualized inflation, computed using the monthly Personal Consumption Expenditures Price Index, Excluding Food and Energy reported by the Bureau of Economic Analysis. This is the so-called core inflation metric favored by the Federal Reserve. The *Real WACC* is the variable that we use in most of our empirical tests. For ease of exposition, henceforth, when we refer to *WACC* we mean the real WACC. Our final sample spans from January 1989 to December 2022.

4. Descriptive Statistics on WACC Estimates

4.1 Descriptive Statistics

Table 2 reports the cross-sectional distribution of WACC estimates for non-overlapping 10-year periods from 1989 to 2022. Within each sample period, for each firm, the WACC estimated as of June-end is averaged across years. The table provides the cross-sectional distributions of these sample period averages. As can be seen, both the value-weighted and equal-weighted WACC show considerable variation over time. The averages increased from 1989 to 2017 and then started declining reaching a low during the COVID pandemic. Figure 1 displays the monthly value-weighted (Figure 1a) and equal-weighted (Figure 1b) WACC alongside NBER recessionary periods, depicted as vertical gray bars. Additionally, the figure includes median and two-standard deviation bands starting in 1997, with the medians and standard deviations computed using data from 1989 onward. The aggregate cost of capital shows considerable time variation across different market cycles and, in general, tends to trough about 6 to 12 months before the onset of a recession. Furthermore, the cost of capital reached a peak in 2010, and by 2022 had fallen to its lowest level in over three decades. Finally, in every period, the EW WACC is greater than the VW WACC signifying a healthy size premium of about 2% to 3% in firm-wide cost of capital.

As discussed in the introduction, there is extensive literature on estimating the implied cost of equity using analyst EPS forecasts. How does the implied WACC compare to the implied cost of equity? To address this question, we first derive the sequential cost of equity from the implied WACC. This is done using the traditional WACC formula, with the ratio of interest expenses to the book value of debt serving as a proxy for the cost of debt—a convenient assumption based on Frank and Shen (2015). We term this the *sequential cost of equity* to distinguish it from the *direct cost of equity*, which is estimated using a free cash flow to equity model with analyst EPS forecasts, following the methodology of Li, Ng, and Swaminathan (2015). For the overall sample, the correlation between the two measures is 59%. This relatively high correlation is encouraging, given that the two approaches rely on distinct datasets and valuation models.

4.2 Industry Cost of Capital

Table 3 reports the cross-sectional distribution of industry costs of capital. The distribution is computed each month for each industry and then averaged across all months. Panel A reports results for 2-digit GICS sectors and Panel B reports results for more homogeneous industries based on 4-digit, 6-digit and 8-digit GICS codes. At the sector level, WACCs are higher for risker sectors. The estimated value-weighted WACC for the Materials (15) and Information Technology (45) sectors, often considered riskier, are 9.69% and 8.62%, respectively. In contrast, the less risky Utilities (55) and Consumer Staples (30) sectors exhibit lower values, at 4.42% and 6.23%, respectively.

At the more homogeneous industry level, the two industries that exhibit the highest WACC are Aluminum, Diversified Metals & Mining, Gold, and Precious Metals & Minerals (11.50%) and Automobiles and Components (11.22%). The two industries with the lowest WACC are both in the Utilities sector – Independent Power and Renewable Electricity Producers (4.01%) and Electric Utilities (4.21%).

There is also a significant size premium in the industry costs o*f capital. For every sector and industry, equal-weighted WACC estimates are consistently higher than their corresponding value-weighted estimates, there is considerable variation in premium across sectors. For instance, in the

Energy sector (10), there is a 3% size premium, while in the Utilities sector (55), the size premium is negligible.

Figure 2 displays the median levered and unlevered WACC (unlevered cost of equity) for GICS 4-digit industries. The unlevered cost of capital is computed using the Modigliani and Miller WACC formula: Unlevered WACC = Levered WACC/(1 – Tax Rate * Debt/Enterprise Value). Not surprisingly, the unlevered WACC is always higher than the levered WACC with the gap between them larger in industries with more leverage.

4.3 Cross-sectional Determinants of WACC

In this section, we examine the cross-sectional determinants of WACC. Our objective is to understand how various firm characteristics such as size, leverage and various measures of risk might be related to cost of capital. To this end, we estimate the following cross-sectional regression each month:

$$\begin{aligned} WACC_t = a &+ b_1 LnSze_{it-1} + b_2\beta_{Mkt,it-1} + b_3\beta_{SMB,it-1} + b_4\beta_{HML,it-1} \\ &+ b_5\beta_{RMW,it-1} + b_6\beta_{CMA,it-1} + b_7 Volatility_{it-1} + b_8 Tax\ Rate_{it-1} \\ &+ b_9 Debt\ Maturity_{it-1} + b_{10} Leverage_{it-1} + e_{it}, \end{aligned} \tag{15}$$

LnSze is the natural log of market cap in millions of dollars. β_{Mkt}, β_{SMB}, β_{HML}, β_{RMW}, *and* β_{CMA} are respectively the Market Beta, SMB Beta, HML Beta, RMW Beta, and CMA Beta estimated from the Fama-French 5-factor model. *Volatility* is the annualized standard deviation monthly stock returns. Both the betas and the standard deviations are estimated using returns from the prior 24 to 60 months. *Idio. Volatility* is the annualized standard deviation of the residual from the Fama-French 5-factor model. *Tax Rate* is the effective tax rate defined earlier, and *Debt Maturity* represents the ratio of long-term interest-bearing debt to short-term interest-bearing debt. *Leverage* is either the *Market Leverage* or the *Book Leverage*. Market Leverage is the ratio of all interest-bearing debt to enterprise value with cash; Book Leverage is the ratio of interest-bearing debt to invested capital with cash.

Table 4 provides descriptive statistics on these firm characteristics. The reported statistics are time-series averages of monthly cross-sectional statistics. For size, we report size ranks determined based on NYSE size cutoffs. The median firm in our sample is a small/midcap firm. Consistent with this, the median firm loads positively on the size factor. The median SMB beta is 0.64. The median effective tax rate in the sample is 32%. The median leverage ratios for both market and book are fairly low, coming in at 16% and 32% respectively. The debt maturity ratio is close to 1, suggesting that the median firm has about the same amount of short-term and long-term debt on its balance sheet.

Table 5 presents the cross-sectional regression (equation 15) results. Specifications (1) to (4) are panel regressions where all independent variables are in Z-scores and standard errors are clustered by firm and year. Specifications (5) to (8) are Fama-MacBeth regressions where the standard errors of the time-series averages are corrected with 12 Newey-West lags. The results show that the cost of capital is negatively related to size, confirming the findings in Table 3. There is a strong positive relationship between the weighted average cost of capital and market beta, SMB beta and HML beta suggesting riskier firms warrant a higher cost of capital. Of particular interest is the investment beta, β_{CMA}, which measures the relative risk of firms that invest conservatively versus those that invest aggressively. This is analogous to the investment beta from the Q-factor model proposed by Hou, Xue, and Zhang (2015). Our results indicate that there is no relationship between the investment beta and the weighted average cost of capital. This suggests that investment risks are not priced in determining an appropriate risk-adjusted firm-wide cost of capital. The profitability beta has a negative sign, but it is also not significant. Thus, Market, SMB, and HML betas emerge as more significant determinants of the cross-section of firm-wide cost of capital compared to the investment beta or the profitability beta.

Both total volatility and idiosyncratic volatility are also significantly positively related to the cost of capital, perhaps due to missing factors. Tax rate and financial leverage (especially book leverage) are negatively related to WACC which suggests that tax advantage of debt due to interest tax shields is an important determinant of the weighted average cost of capital. Debt maturity is

also weakly negatively related to cost of capital which suggests firms with longer maturity debt face a lower cost of capital which is consistent with the *rollover risk* hypothesis.[15]

In the rank regressions, the slope coefficients are interpretable as cost of capital/expected return premiums. The two firm characteristics with the largest premiums are volatility (both total and idiosyncratic) and tax rate with premiums in the range of 3% to 4% in absolute values. The market, SMB, and HML betas exhibit premiums in the range of 0.6% to 1.7%. In summary, the cross-sectional results affirm the usefulness and applicability of our direct WACC methodology.

5. Corporate Investment and WACC at the Firm Level

The standard Q-theory model developed by Frank and Shen (2016) predicts a negative relationship between cost of capital and corporate investment, after controlling for profitability. We test this relationship using our direct measure of WACC. Following Frank and Shen (2016), we estimate panel regressions in which we regress a firm's future investment on contemporaneous Cash Flow and past WACC. *Investment* is defined as the ratio of (Capital expenditures + Cash acquisitions – Net sale of property) and average net property plant equipment over the measurement period. We measure investment over two periods: the next quarter *(t+1)* and the next four quarters *(t+1, t+4)*. *Cash Flow* is defined as the next quarter's (t+1) or the next four quarters' (t+1 to t+4) Operating Income Before Depreciation and Amortization (EBITDA), divided by the average net property, plant, and equipment over that period.[16] WACC is measured as of the end of quarter *t*.

In line with Frank and Shen (2016), we exclude utility firms from our analysis on top of excluding the financials. Additionally, to mitigate the impact of outliers on our regressions, we remove the top and bottom 0.5% of all variables. In the panel regressions, we include firm and year fixed effects to account for potential omitted variables. Standard errors are computed by clustering at

[15] Rollover risk hypothesis suggests that short-term debts intensify the shareholder-debtholder conflict, in which shareholders are motivated to default earlier leading to greater credit risk (see He and Xiong, 2012).

[16] Lewellen and Lewellen (2016) critique the traditional measure of cash flows used in the literature, which is typically calculated as the sum of net income before extraordinary items and depreciation and amortization, due to the inclusion of various accrued and non-cash items in income. They propose using cash flow from operations (CFO) from the statement of cash flows instead. Since our goal is to estimate firm-wide cost of capital, we follow Frank and Shen (2016) and rely on EBITDA, a widely used measure of operating cash flow that, unlike CFO, is calculated before interest expenses.

both the firm and year levels to address potential heteroskedasticity and correlation within each cluster.

Table 4 presents descriptive statistics for Investment and Cash Flow. The median quarterly investment for the firm in our sample is 6% and the median four-quarter investment is about four times that at 27%. The median quarterly cash flow and four quarter cash flow (as a percent of NPPE) are 16% and 64% respectively. The results of the panel regression are presented in Table 6. Both the next quarter investment and the next four quarter investment are negatively related to the current WACC controlling for contemporaneous cash flows. The relationship is the strongest for the four-quarter investment. This makes sense since changes in cost of capital should take time to be fully reflected in corporate investment policy. Not surprisingly, investment is also strongly positively related to contemporaneous cash flows.

6. Investment, Cost of Capital and Stock Returns

Firms that invest aggressively underperform firms that invest conservatively, and profitable firms outperform less profitable firms.[17] Based on this finding, Hou, Xue and Zhang (2015) augment the Fama and French 3-factor model with an investment factor and a profitability factor and refer to the resulting 5-factor model as the *q*-factor model. Subsequently, Fama and French (2015) expanded on their earlier 3-factor model to construct a 5-factor model by incorporating their own investment (Conservative Minus Aggressive, CMA) and profitability (Robust Minus Weak, RMW) factors.

Why do high investment firms underperform low investment firms? Hou, Xue, and Xhang (2015) provide an explanation based on their investment asset pricing model motivated by Q-theory. Their model predicts that for a given profitability, high investment firms should earn lower expected returns than do low investment firms. This is because low cost of capital implies high marginal Q and high investment, and high cost of capital implies low marginal Q and low investment. Intuitively, holding profitability constant, a lower hurdle rate would lead to more investment and a higher hurdle rate would lead to less investment. This is just textbook capital budgeting. By

[17] See Titman, Wei, and Xie (2004), Hirshleifer, Hou, Teoh, and Zhang (2004), and Cooper, Gulen, and Schill (2008), Soliman (2008) and Novy-Marx (2013).

investing, high-investment firms exercise their riskier growth options, converting them into less risky assets-in-place (see Berk, Green, and Naik, 1999), thereby becoming less risky overall. Thus, high investment firms tend to underperform because they are less risky.

However, when controlling for cost of capital, the expected relationship between investment levels and future returns presents an intriguing dynamic. Firms with a high cost of capital face a higher hurdle rate, necessitating substantial returns on investment to justify their high capital expenditures. Consequently, firms that invest heavily despite high costs of capital can only do so with the expectation of achieving high returns on capital that surpass their elevated cost of capital. This implies that such firms should earn high future stock returns, rather than low returns. This aligns with a risk-based interpretation, suggesting that only firms confident in their profitable opportunities would undertake significant investments despite elevated capital costs.

Conversely, high-investment firms with a low cost of capital are expected to generate lower future returns, reflecting their reduced inherent risk as predicted by Hou, Xue, and Zhang (2015). Low-investment firms with a low cost of capital present a conundrum, however. Why do they invest less in spite of their low cost of capital? They may invest less due to limited availability of positive NPV projects, even when facing low hurdle rates. This suggests that, despite having favorable financing conditions, these firms might lack sufficiently profitable opportunities to pursue. This discussion underscores the importance of considering cost of capital as a critical lens for evaluating investment behavior and expected returns.

Table 7 presents the results of testing these hypotheses. We construct univariate and bivariate portfolios monthly from 1989 to 2022, sorting based on investment and WACC and report their average monthly returns and Fama and French 5-factor intercepts. Panel A displays univariate results for 5 portfolios, while Panels B through G present bivariate results for 25 portfolios, based on an independent 2-way sort of WACC and investment.

The results in Panel A show that, as expected, the high-investment portfolio underperforms the low-investment portfolio in terms of excess returns (in excess of T-bill returns) but not on a risk-adjusted basis. Similarly, for WACC portfolios, the High-WACC portfolio outperforms the Low-

WACC portfolio in excess returns; however, after accounting for risk differences, there is no significant outperformance. The more intriguing findings appear in Panels B to G which show that conditional on cost of capital, high investment firms do not uniformly underperform low investment firms. The excess returns in Panel B indicate that high-investment portfolios underperform low-investment portfolios more strongly among firms with higher costs of capital, with the strongest underperformance observed among High-WACC firms. Specifically, the High WACC-High Investment portfolio underperforms the High WACC-Low Investment portfolio by a statistically significant 0.63% per month. This underperformance decreases to 0.41% (Panel C) after risk adjustment but remains statistically significant.

Turning to profitability and growth, Panels D and E reveal that these are high-growth, highly profitable firms. The median EBITDA/NPPE ratio is 135% (Panel D), and the expected growth rate is 59% (Panel E), the highest among all 25 portfolios. However, despite their high profitability, these firms are characterized by negative RMW betas, indicating that their returns covary more with low-profitability firms than with high-profitability firms (see Panel H). These firms also have a high reinvestment rate of 0.76 (Panel F). As expected, their investment beta is negative (Panel G), reflecting their status as high-investment firms.

As hypothesized, High Investment-High WACC firms, which are aggressive investors despite their high cost of capital, seem to be doing so under the expectation of substantial profitability from their investments. There is no apparent risk-based or profitability-based reason to predict their poor stock performance. Yet, their shares underperform in both excess and risk-adjusted terms, inconsistent with a risk explanation. One potential reason for this underperformance could be overinvestment, as these firms may channel resources into projects with negative marginal Q or NPV due to managerial overoptimism about future profitability, which may also influence investors. In other words, given their high cost of capital, they may not be profitable enough.

In contrast, among Low WACC firms, the High Investment portfolio outperforms the Low Investment portfolio on a risk-adjusted basis by a statistically significant 0.31% per month. As expected, these firms are also characterized by a negative investment beta, indicative of aggressive investment behavior. Despite being low-growth firms, they are highly profitable, with an expected

growth rate of 5% and profitability of 117%. Their profitability is slightly lower than that of High-WACC-High Investment firms but, given their low cost of capital, perhaps high enough to generate positive NPVs for their projects. Their outperformance contrasts starkly with the conventional wisdom that high-investment firms underperform due to lower risk and, consequently, lower expected returns. These results indicate that whether high-investment firms underperform or outperform depends on the interaction between their cost of capital and profitability. As hypothesized, the Low WACC-Low Investment firms exhibit the lowest growth expectations and profitability, with a growth rate of 2% and profitability of 24% justifying their low investment, which may explain their underperformance relative to Low WACC-High Investment firms. Despite their low growth and profitability, they still invest more than other Low Investment firms perhaps because of their lower cost of capital, with a reinvestment rate of 0.42 compared to 0.15 for High WACC-Low Investment firms.

The proxy for investment used in Table 7 captures growth in property, plant, and equipment but does not account for growth in other balance sheet assets such as cash, non-cash current assets, and other long-term assets. To address this limitation, we repeat the analysis in Table 7 using a broader measure of growth—growth in total assets, as studied by Cooper, Gulen, and Schill (2008). Total asset growth is computed over the trailing four quarters. The results based on this expanded measure are presented in Table 8.

The results in Table 8 broadly mirror the findings in Table 7. The underperformance of high asset growth firms increases monotonically with WACC, with the largest underperformance observed in the High WACC portfolio and the smallest in the Low WACC portfolio.[18] However, unlike Table 7, high asset growth, low WACC firms do not outperform on a risk-adjusted basis (see Panel C), instead earning zero abnormal returns. Despite this, the fundamental conclusion holds; firms with the highest asset growth and highest WACC underperform despite their high cost of capital, growth, and profitability—an outcome inconsistent with a risk-based explanation. Overinvestment, potentially driven by over-optimistic expectations regarding future growth and

[18] Bivariate portfolios based on direct ICC (implied cost of equity) and investment produce substantially similar results. The underperformance of high investment firms is concentrated among those with high implied cost of equity.

profitability, may be a key factor. These firms also exhibit the highest growth expectations and very high reinvestment rates, with a negative CMA beta confirming their status as aggressive investors. Conversely, Low WACC, Low Investment firms have the lowest growth expectations and profitability, yet they invest more than other Low Investment firms most likely because of their lower cost of capital.

The key takeaways are as follows: Not all high-investment firms uniformly underperform relative to low-investment firms. Conditioning on cost of capital reveals distinct performance patterns: High WACC-High Investment firms underperform, while Low WACC-High Investment firms either outperform or perform neutrally. The High WACC-High Investment firms appear to exhibit clear signs of overinvestment. Our results indicate that the underperformance of high-investment firms, at a minimum, necessitates a more nuanced interpretation of the balance between risk and reward.

7. Aggregate WACC and Corporate Investment

Abel and Blanchard (1986) investigate the relationship between aggregate corporate investment and marginal Q and find that they are positively related. They also find that the fluctuations in marginal Q are driven mostly by fluctuations in the cost of capital rather than marginal profits. When the two components of marginal Q are included separately in the investment regression, they find that the marginal profit component had a stronger effect on investment. However, due to the lack of reliable measures of cost of capital, they do not directly test the relationship between aggregate investment and aggregate cost of capital. We fill this gap by directly examining the relationship between aggregate investment and aggregate WACC.

Kothari, Lewellen, and Warner (2016) find that aggregate corporate investment is largely driven by past profits and stock returns, with minimal connection to interest rates, the term spread, or the default spread. Figure 3 plots investment, WACC, term spread and default spread. There appears to be an inverse relationship between investment and WACC. For instance, investment peaked and WACC bottomed out in early 2000. In contrast, investment and WACC peaked during the 2008/2009 global financial crisis. There are other similar episodes displayed in Figure 3. The pandemic was an exception when both cost of capital and investment nosedived.

We now formally test the relationship between future aggregate investment and past WACC using quarterly data from 1989 to 2022. In the regressions, we include a pandemic dummy variable which takes on a value of 1 for the period 2020-Q1 to 2022-Q4 and zero otherwise. This takes into account the anomalous occurrence when both investment and cost of capital fell during the pandemic. We test the relationship between this quarter's WACC and the next quarter's investment as well as the next four-quarter investment. Panel A of Table 8 presents the results based on next quarter's investment and Panel B on the next four-quarter investment. The t-statistics in parentheses are based on Newey-West correction with 12 lags. The aggregate WACC is computed by value-weighting the individual WACCs by prior quarter's enterprise value.

The results reveal a strong negative relationship between WACC and future investment, even when accounting for cash flows. This significant finding addresses a challenge that has long eluded empirical tests of Q-theory—specifically, understanding the economic determinants of the cyclical variability in investment. As expected, corporate business investment is strongly influenced by fluctuations in the aggregate cost of capital. We also confirm the strong positive relationship between profitability and future investment as reported by Kothari, Lewellen and Warner (2016). Next, we include past term spread and default spread in the investment regression. Term spread is the monthly yield on AAA bonds minus the monthly yield on 3-month T-bill. Default spread is the monthly yield on BAA bonds minus the monthly yield on AAA bonds. All interest rates are available in FRED (Federal Reserve of St. Louis Research Database). Figure 3 also displays the time-series evolution of term spread and default spread. Term spread, in particular, seems to comove quite strongly with WACC. Indeed, the time-series correlation between the two is a fairly high 61%.

We find that term spread is also significantly negatively related to future investment. Moreover, including the term spread reduces the significance of WACC. This suggests that interest rate dynamics play a more substantial role in shaping future investment decisions than previously recognized. The default spread is also negatively related to future investment and is significant in predicting the next four-quarter investment.

Our multiple regression results indicate that lagged market returns predict lower investment in the subsequent quarter (see Panel A). This contrasts with the findings of Kothari, Lewellen, and Warner (2016), who observed that high stock returns predict increased investment during the 1959-2010 period. However, the predictive power of past market returns diminishes when forecasting investment over the next four quarters. These differences may reflect unique characteristics of the sample period we examined.

8. Conclusions

In this paper, we develop a new methodology for computing the weighted average cost of capital (WACC) by incorporating forward-looking expectations of operating profits. This approach eliminates the need to estimate individual components of the traditional WACC formula, such as the cost of equity and the cost of debt. The resulting WACC estimates vary cross-sectionally as expected across industries, consistent with their respective risk profiles, and exhibit relationships in the cross-section with various risk measures, leverage, tax rates, and debt maturity, as anticipated. Our results indicate that firms with a high cost of capital tend to invest less in the future.

Our findings reveal that not all high-investment firms uniformly underperform low-investment firms; rather, their performance is contingent on their cost of capital. High WACC-High Investment firms tend to underperform, while Low WACC-High Investment firms outperform. This pattern suggests that overinvestment, driven by either high hurdle rates or low growth expectations, may contribute to observed returns. These results highlight the need for a nuanced interpretation of the relationship between investment behavior and expected returns, considering both risk and profitability dynamics.

At the macro level, an increase in the aggregate cost of capital leads to lower future investment; however, this relationship is moderated by interest rate dynamics. Specifically, in the presence of the term spread, the impact of investment becomes less significant, while the term spread itself is significantly negatively related to future investment. Overall, our approach offers corporate managers a straightforward method for estimating firm-wide and divisional cost of capital and provides researchers with a more reliable measure of the weighted average cost of capital.

Appendix: Compustat Items

EBITDA (OIBDPQ)	: Operating Income Before Depreciation
EBIT (OIADPQ)	: Operating Income After Depreciation
$EPS_{OP,0}$ (OEPS12)	: Earnings Per Share from Operations - 12 Months Moving
Ibq	: Net income before extraordinary items
$Spiq$	: Special items.
$ShrsOut$ (CSH12Q)	: Common Shares Used to Calculate Earnings Per Share - 12 Months Moving Average
$CapEx$ (CAPXY)	: Capital Expenditures
$CashAcq$ (AQCY)	: Acquisitions
Dep&Amort (DPCY)	: Depreciation and Amortization
ΔRec (-RECCHY)	: Accounts Receivable - Decrease (Increase)
ΔInv (-INVCHY)	: Inventory - Decrease (Increase)
$\Delta AcctPay$ (APALCHY)	: Accounts Payable and Accrued Liabilities - Increase (Decrease)
TXTQ	: Income Taxes – Total
PIQ	: Pretax Income
PSTKRQ	: Preferred/Preference Stock – Redeemable
CHEQ	: Cash and Short-Term Investments
MIBQ	: Noncontrolling Interest - Redeemable
DLCQ	: Debt in Current Liabilities
DLTTQ	: Long-Term Debt - Total

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Table 1
Summary Statistics

This table presents summary statistics based on annual data as of June of each year from 1989 to 2022. *Listed Firms* is the number of publicly traded firms in NYSE/AMEX/NASDAQ (share codes 10 and 11 only) from CRSP. *IBES EPS Forecasts*, *IBES EBIT Forecasts*, and *IBES EBITDA Forecasts* are respectively the number of *non-financial* firms with available FY1 (Fiscal Year 1) EPS (Earnings Per Share), EBIT (Earnings Before Interest and Taxes), and EBITDA (Earnings Before Interest, Depreciation and Amortization) forecasts in the IBES database. *EPS Forecasts*, *EBIT Forecasts* and *EBITDA Forecasts* represent firms in our sample with positive FY1 forecasts and for which we are able to compute Weighted Average Cost of Capital (WACC). In this sample, we drop the largest 0.5% and smallest 0.5% of WACC estimates (each month) to eliminate outiers. *EPS Analysts*, *EBIT Analysts* and *EBITDA Analysts* are the median number of analysts for the firms in our sample in June of each year.

Year	Listed Firms	IBES EPS Forecasts	IBES EBIT Forecasts	IBES EBITDA Forecasts	EPS Forecasts	EBIT Forecasts	EBITDA Forecasts	EPS Analysts	EBIT Analysts	EBITDA Analysts
1989	5,877	2,326	0	0	1,452	0	0	6	0	0
1990	5,763	2,313	0	0	1,634	0	0	6	0	0
1991	5,629	2,307	0	0	1,686	0	0	6	0	0
1992	5,784	2,527	0	0	1,836	0	0	5	0	0
1993	6,003	2,801	0	0	2,078	0	0	5	0	0
1994	6,591	3,195	0	0	2,366	0	0	5	0	0
1995	6,689	3,312	0	0	2,472	0	0	5	0	0
1996	7,112	3,744	0	0	2,630	0	0	5	0	0
1997	7,379	4,048	0	0	2,807	0	0	5	0	0
1998	7,255	4,045	0	0	2,785	0	0	5	0	0
1999	6,706	3,773	0	104	2,504	0	65	5	0	1
2000	6,515	3,720	0	245	2,178	0	99	5	0	1
2001	5,945	3,245	6	164	1,755	3	66	6	2	1
2002	5,411	2,763	330	867	1,695	264	668	6	1	1
2003	4,984	2,599	1,074	1,711	1,771	865	1,328	6	1	2
2004	4,815	2,717	1,215	1,907	1,991	1,050	1,574	6	2	2
2005	4,759	2,782	1,301	2,076	2,038	1,107	1,677	6	2	3
2006	4,695	2,834	1,425	2,251	2,115	1,232	1,834	6	2	3
2007	4,622	2,845	1,341	2,276	2,061	1,136	1,797	7	1	3
2008	4,497	2,750	1,411	2,283	1,976	1,185	1,785	6	2	3
2009	4,155	2,572	1,403	2,043	1,700	1,094	1,503	7	2	4
2010	4,017	2,593	1,574	2,165	1,956	1,330	1,769	7	3	4
2011	3,835	2,481	1,708	2,148	1,936	1,449	1,781	7	3	5
2012	3,730	2,490	2,284	2,158	1,975	1,849	1,807	8	6	6
2013	3,614	2,419	2,289	2,129	1,878	1,798	1,743	8	7	6
2014	3,702	2,527	2,409	2,244	1,871	1,793	1,751	8	7	6
2015	3,745	2,546	2,471	2,279	1,796	1,748	1,711	8	6	6
2016	3,655	2,498	2,432	2,221	1,705	1,668	1,635	7	6	6
2017	3,579	2,433	2,364	2,130	1,721	1,667	1,631	7	6	6
2018	3,570	2,440	2,352	2,151	1,711	1,651	1,610	7	6	6
2019	3,597	2,523	2,449	2,197	1,707	1,664	1,601	7	6	6
2020	3,557	2,495	2,406	2,154	1,418	1,369	1,328	7	5	6
2021	4,051	2,742	2,679	2,374	1,665	1,619	1,572	7	5	6
2022	4,427	3,001	2,916	2,567	1,704	1,655	1,604	7	6	6

Table 2
Summary Statistics of Real WACC

This table presents summary statistics for real WACC (in percent), using annual data as of June of each year, for various sample periods from 1989 to 2022. Real WACC is estimated using the Fisher model as (Nominal WACC - Inflation)/(1+ Inflation). Inflation is estimated as a 12 month average of annualized monthly inflation based on the price index for Personal Cosumption Expenditures excluding the more volatile food and energy prices. *EW Mean* is the equal-weighted mean and *VW Mean* is the value-weighted mean with weights based on month-end (same as for WACC estimation) enterpise value. *NFRMS* is the average number of firms per year.

Year	5%	25%	Median	EW Mean	VW Mean	StdDev	75%	95%	NFRMS
1989-1997	0.8	4.7	8.2	10.2	6.9	8.5	13.2	27.0	2,175
1998-2007	1.2	5.0	7.9	9.8	6.7	8.0	12.3	25.5	2,010
2008-2017	2.2	5.3	7.9	10.0	7.8	7.6	12.2	25.3	1,827
2018-2022	0.5	3.0	5.6	8.1	5.3	8.6	10.3	23.5	1,627
2022	-0.9	1.5	4.3	7.7	3.7	11.5	9.5	25.4	1,707

Table 3
Summary Statistics of Industry WACC

This table presents summary statistics for real WACC (in percent) for various industries. Panel A presents results for industries defined on 2-digit GICS codes and Panel B presents results for more homogeneous industries constructed from 4 digit, 6-digit, or 8-digit GICS codes. The statistics are computed each month for each industry and then averaged across all months. *EW Mean* represents the equal-weighted average and *VW Mean* represents the value-weighted average, with weights based on end of month enterprise value. *NFRMS* is the average of number of firms.

Panel A: Industry WACC Based on 2-digit GICS

Industry	GICS Codes	5%	25%	Median	EW Mean	VW Mean	StdDev	75%	95%	NFRMS
Energy	10	0.43	4.20	8.08	10.38	7.31	27.86	8.98	14.34	101
Materials	15	1.93	5.37	8.32	10.72	9.69	28.00	8.55	13.41	137
Industrials	20	1.72	5.21	8.10	10.35	7.06	27.09	8.34	12.85	463
Consumer Discretionary	25	1.94	5.57	8.18	10.04	6.94	24.66	7.51	12.32	381
Consumer Staples	30	1.82	4.40	6.19	7.74	6.23	19.82	6.14	9.00	104
Health Care	35	1.10	4.79	7.37	9.27	6.47	24.06	7.59	11.52	182
Information Technology	45	0.96	5.41	9.21	11.30	8.62	28.84	8.92	14.93	256
Communication Services	50	0.04	3.99	7.12	9.16	6.75	25.58	8.40	11.81	195
Utilities	55	1.28	2.64	3.74	4.92	4.42	13.20	4.40	5.57	108

Panel B: Industry WACC based on more homoegenous GICS (4, 6, or 8 digits)

Industry	GICS Codes	5%	25%	Median	EW Mean	VW Mean	StdDev	75%	95%	NFRMS
Energy, Equipment and Services	101010	1.64	5.58	9.65	11.16	10.48	27.90	8.06	15.13	15
Integrated Oil & Gas, Oil & Gas Refining and Marketing, Oil & Gas Storage and Transportation, Coal and Consumeable Fuels	10102010, 10102030, 10102040, 10102050	1.10	4.79	8.64	11.07	6.84	32.26	9.23	15.12	17
Oil & Gas Exploration and Production	10102020	0.34	3.98	7.96	10.01	7.66	26.74	8.65	13.93	69
Commodity, Diversified, and Fertilizers & Agricultural Chemicals	15101010, 15101020, 15101030	2.41	5.57	8.15	10.14	8.38	24.86	7.56	12.40	26
Specialty Chemicals	15101050	2.26	5.53	8.05	10.03	9.44	26.32	7.44	12.29	21
Construction Materials, Containers and Packaging	151020, 151030	2.07	5.15	7.75	9.72	8.08	27.51	7.40	11.98	20
Aluminum, Diversified Metals & Mining, Gold, and Precious Metals & Minerals	15104010, 15104020, 15104030, 15104040	1.28	5.21	9.44	11.71	11.50	32.39	9.35	15.88	19
Steel	15104050	2.22	6.00	10.16	12.04	11.17	28.92	8.52	16.07	28
Paper and Forest Products	151050	2.42	5.23	8.08	10.35	9.90	26.63	7.83	13.49	23
Aerospace and Defense	201010	2.20	5.33	7.69	9.26	7.92	23.37	6.64	11.06	39
Building Products	201020	2.05	5.60	8.71	10.45	8.91	26.15	7.56	13.15	45
Construction and Engineering	201030	1.81	5.07	7.92	9.75	7.44	24.70	7.44	12.22	62
Electrical Equipment	201040	2.24	5.77	8.75	10.82	8.86	29.22	7.90	13.50	20
Industrial Conglomerates	201050	3.83	7.10	10.32	12.79	9.15	31.01	9.84	15.76	148
Construction Machinery and Heavy Transportation Equipment	20106010	2.06	5.46	8.69	10.66	6.01	27.72	7.96	13.51	40
Industrial Machinery, Supplies and Components	20106020	1.84	4.84	7.89	10.22	7.35	26.92	8.21	13.10	81
Trading Companies and Distributors	201070	1.80	5.28	7.85	9.72	8.58	27.42	7.36	12.25	13
Commercial Services and Supplies	202010	2.06	4.92	7.40	9.06	6.69	24.05	6.46	11.46	14
Professional Services	202020	2.56	6.18	9.46	10.94	7.51	26.30	6.99	14.41	17
Air Freight and Logistics, Marine Transportation and Transportation Infrastructure	203010, 203030, 203050	1.63	4.81	7.33	8.91	8.28	23.98	6.41	11.38	14
Passenger Airlines	203020	1.63	5.85	9.46	11.56	9.79	30.43	8.60	15.46	15
Ground Transportation	203040	2.10	5.17	7.23	8.85	8.03	21.57	6.51	10.58	32
Automobiles and Components	2510	3.03	6.63	10.13	11.91	11.22	27.67	8.04	15.04	27
Home Furnishings and Homebuilding	25201020, 25201030	2.10	5.98	9.51	11.11	9.25	26.42	7.70	14.38	31

Industry	GICS Codes	5%	25%	Median	EW Mean	VW Mean	StdDev	75%	95%	NFRMS
Consumer Electronics and Household Appliance	25201010, 25201040	2.13	5.75	8.62	10.59	7.78	28.21	7.63	13.41	18
Leisure Products	252020	2.71	6.05	8.92	10.79	7.04	26.29	7.40	13.57	22
Apparel Accessories & Luxury goods	25203010	2.63	6.16	9.09	10.80	8.37	27.94	7.34	13.47	18
Footwear and Textiles	25203020, 25203030	2.87	7.01	9.85	11.77	8.90	29.23	7.80	14.39	21
Casinos and Gaming	25301010	2.02	5.39	8.77	10.84	8.03	28.18	8.37	13.97	17
Hotels, Resort and Cruise Lines	25301020	1.71	4.76	7.11	8.86	7.53	25.03	6.82	10.85	15
Diversified Consumer Services	253020	1.36	5.04	7.64	9.47	7.03	25.59	7.33	11.83	27
Distributors, Diversified Consumer Services	255010, 255020	1.87	5.72	8.26	9.88	7.50	24.13	6.62	12.83	16
Broadline Retail	255030	2.46	5.08	6.95	8.14	5.68	18.79	5.13	9.87	21
Apparel Retail	25504010	3.60	7.15	9.52	10.99	8.85	24.15	6.46	13.12	34
Other Specialty Retail	25504040	1.61	5.52	8.03	9.75	7.87	24.73	7.24	12.05	42
Computer and Electronics Retail, and Automotive Retail	25504020, 25504050	1.37	4.22	6.79	8.30	5.81	21.83	6.38	10.58	19
Home Improvement Retail and Homefurnishing Retail	25504030, 25504060	2.55	5.66	7.50	9.01	6.22	22.41	5.54	11.10	16
Consumer Staples Distribution and Retail	301010	1.55	4.29	6.14	7.49	6.57	19.65	6.00	8.59	30
Beverages and Tobaco	302010, 302030	1.80	4.99	6.76	8.05	6.57	21.68	5.33	9.65	17
Food Products	302020	1.98	4.31	6.21	8.06	6.13	21.13	6.35	9.66	44
Household Products	303010	1.94	4.52	5.84	6.74	5.48	17.24	4.09	7.85	13
Health Care Equipment and Supplies	351010	1.43	4.94	7.30	8.85	6.55	22.11	6.71	10.84	61
Health Care Providers and Services	351020	1.46	5.02	7.36	9.06	7.35	22.53	6.90	11.14	61
Pharmaceuticals, Biotechnology and Life Sciences	3520	0.77	4.13	7.42	9.66	6.38	26.37	8.38	12.76	60
Software	451030	0.57	4.79	8.70	10.48	8.07	27.19	8.47	13.95	55
Communications Equipment	452010	0.94	5.20	9.63	11.80	8.71	31.21	9.45	16.06	38
Technology Hardware, Storage and Peripherals	452020	1.54	6.03	9.70	11.75	8.93	30.18	8.74	15.50	31
Electronic Equipment and Instruments & Electronic Components	45203010, 45203015	1.75	6.07	9.66	11.67	8.68	29.02	8.41	15.61	24
Electronic Manufacturing Services	45203020	1.68	5.79	9.47	11.37	8.62	28.41	8.49	14.74	38
Technology Distributors	45203030	2.42	5.62	8.24	9.99	8.86	25.53	6.90	12.48	13
Semiconductor Materials and Equipment	45301010	1.31	5.82	10.75	12.92	9.92	33.34	9.96	18.07	21
Semiconductors	45301020	1.06	5.54	9.72	11.34	9.20	28.07	8.40	15.35	36
Diversified Telecommunication Services	501010	-0.09	3.12	5.42	7.59	5.71	24.36	7.81	9.36	32
Media and Entertainment	502010, 502020	-0.11	3.69	7.00	8.80	7.19	25.27	8.15	11.34	68
Interactive Media and Services	502030	0.98	4.84	7.96	10.04	7.76	26.25	8.25	13.00	95
Electric Utilities	551010	1.70	2.86	3.76	4.55	4.21	10.77	3.33	5.02	40
Gas Utilities	551020	1.61	2.96	4.18	5.25	5.12	12.97	4.00	6.16	29
Water Utilities	551040	0.90	2.80	4.50	5.99	5.41	18.37	5.27	7.29	21
Independent Power and Renewable Electricity Producers	551050	1.02	2.48	3.41	4.25	4.01	12.42	3.28	4.88	19

Table 4
Cross-Sectional Summary Statistics

This table presents time-series averages of monthly cross-sectional summary statistics for the firms in our sample, covering the period from January 1989 to December 2022. *SzRnk* represents the rank of individual stocks from 1 (Small) to 10 (Large), based on NYSE market capitalization cutoffs. *Market Beta, SMB Beta, HML Beta, RMW Beta,* and *CMA Beta* refer to the market, size, value, profitability, and investment betas, respectively, derived from the Fama-French 5-factor model. *Volatility* and *Idio.Volatility* are the annualized overall and idiosyncratic standard deviations, respectively, of monthly stock returns. *Tax Rate* is the effective tax rate, while *Debt Maturity* represents the ratio of long-term debt to short-term debt. *Market Leverage* is the ratio of all interest-bearing debt to enterprise value with cash; *Book Leverage* is the ratio of interest-bearing debt to invested capital with cash. *Cash Flow* is defined as the next quarter's (t+1) or sum of the next four quarters' (t+1 to t+4) Operating Income Before Depreciation and Amortization (EBITDA), divided by the average net property, plant, and equipment over that period. *Investment* is the sum of capital expenditures and cash acquisitions, less the net sale of property, over the next quarter (t+1) or the next four quarters (t+1 to t+4), divided by the average net property, plant, and equipment over the same period. Betas and volatilities are estimated using trailing 24 to 60 months of stock returns, with Idio.Volatility representing the residual standard error from the 5-factor model.

Variable	5%	25%	Median	Mean	StdDev	75%	95%	NOBS
SzRnk	1.00	2.37	4.51	4.91	2.86	7.27	10.00	1,908
Market Beta	0.15	0.66	1.02	1.06	0.61	1.41	2.15	1,680
SMB Beta	-0.52	0.12	0.64	1.23	0.72	0.87	2.28	1,680
HML Beta	-1.72	-0.53	0.07	0.05	1.06	0.65	1.75	1,680
RMW Beta	-2.20	-0.61	0.13	0.06	1.30	0.81	2.09	1,680
CMA Beta	-2.56	-0.86	-0.01	-0.05	1.48	0.79	2.31	1,680
Volatility	0.20	0.29	0.39	0.42	0.17	0.52	0.75	1,680
Idio. Volatility	0.16	0.23	0.32	0.35	0.15	0.43	0.63	1,680
Tax Rate	0.10	0.18	0.32	0.29	0.13	0.37	0.46	1,928
Debt Maturity	0.10	0.73	0.91	0.80	0.27	0.98	1.00	1,690
Market Leverage	0.00	0.03	0.16	0.21	0.20	0.34	0.61	1,918
Book Leverage	0.00	0.09	0.32	0.34	0.28	0.51	0.81	1,917
Cash Flow (Qtr)	0.00	0.08	0.16	0.30	0.49	0.34	1.06	1,582
Cash Flow (4 Qtr)	0.02	0.31	0.64	1.09	1.65	1.28	3.75	1,476
Investment (Qtr)	0.01	0.03	0.06	0.13	0.32	0.11	0.43	1,582
Investment (4 Qtr)	0.06	0.16	0.27	0.56	1.00	0.52	2.04	1,476

Table 5
Cross-sectional Regression of Real WACC on Firm Characteristics

This table presents cross-sectional regressions of WACC (in percent) in month *t* on various firm characteristics in month *t-1* including size, betas, volatility, idiosyncratic volatility, book & market leverage, tax rate and debt maturity using data from January 1989 to December 2022. Specifications (1), (2), (3) and (4) report panel regressions (pooled time-series and cross-section) where all firm characteristics are converted to Z-scores each month, and Specifications (5), (6), (7) and (8) report monthly Fama-MacBeth regressions where all firm characteristics are converted to cross-sectional ranks (0 to 1) each month. Thus, slope coefficients in (1) to (4) represent the change in WACC for a one-standard deviation change in the firm characteristic while those in (5) to (8) are expected return premiums. All independent variables are as defined in Table 4. t-statistics in parentheses are computed using clustered (by firm and year) standard errors in the panel regressions or Newey-West corrections with 12 lags (Fama-MacBeth).

	Scaled to Z-Scores				**Scaled to Ranks**			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
LnSze	-1.287	-1.260	-1.218	-1.190	-3.991	-3.893	-3.834	-3.747
	(-21.29)	(-20.37)	(-20.66)	(-19.76)	(-17.58)	(-17.32)	(-13.64)	(-13.64)
Market Beta	0.300	0.445	0.276	0.418	1.054	1.802	0.981	1.705
	(6.32)	(10.01)	(5.87)	(9.53)	(4.08)	(7.73)	(3.68)	(7.45)
SMB Beta	0.192	0.281	0.198	0.284	0.673	1.058	0.651	1.023
	(3.97)	(6.01)	(4.07)	(6.07)	(2.84)	(4.67)	(2.76)	(4.52)
HML Beta	0.347	0.33	0.323	0.309	1.257	1.222	1.189	1.175
	(7.36)	(7.04)	(6.87)	(6.60)	(3.91)	(3.62)	(4.15)	(3.90)
RMW Beta	-0.046	-0.101	-0.036	-0.089	-0.218	-0.409	-0.217	-0.402
	(-1.06)	(-2.36)	(-0.82)	(-2.09)	(-1.37)	(-2.33)	(-1.35)	(-2.29)
CMA Beta	0.078	0.033	0.063	0.020	0.303	0.131	0.226	0.077
	(1.64)	(0.69)	(1.33)	(0.42)	(1.55)	(0.67)	(1.26)	(0.44)
Volatility	0.812		0.807		3.784		3.596	
	(12.99)		(13.00)		(15.45)		(14.02)	
Idio.Volatility		0.720		0.720		3.311		3.137
		(12.43)		(12.54)		(15.12)		(14.61)
Tax Rate	-1.124	-1.130	-1.119	-1.125	-3.514	-3.534	-3.493	-3.511
	(-28.19)	(-28.37)	(-28.06)	(-28.22)	(-20.79)	(-20.58)	(-20.88)	(-20.67)
Debt Maturity	-0.046	-0.049	-0.029	-0.030	-0.353	-0.369	-0.251	-0.265
	(-1.12)	(-1.19)	(-0.71)	(-0.72)	(-3.19)	(-3.34)	(-2.24)	(-2.38)
Book Leverage			-0.322	-0.326			-1.184	-1.166
			(-6.47)	(-6.55)			(-3.53)	(-3.49)
Market Leverage	-0.249	-0.246			-0.532	-0.505		
	(-4.54)	(-4.47)			(-1.27)	(-1.20)		
NOBS	570,849	570,849	569,957	569,957	1,403	1,403	1,400	1,400
Adj. R^2 (%)	9.4%	9.4%	9.4%	9.3%	12.1%	12.1%	12.0%	12.0%

Table 6
Cross-Sectional Regression of Investment on Real WACC

This table presents panel regressions with firm and year fixed effects of a firm's investment on its lagged WACC using quarterly data from 1989-Q1 to 2022-Q4. The dependent variable is Investment defined as the next quarter (t+1) capital expenditures plus cash acquisition less net sale of property divided by the average net property plant and equipment over the next quarter (t to t+1) or the sum of next four-quarter (t+1 to t+4) capital expenditures plus cash acquisition less net sale of property divided by the average net property plant and equipment over the four quarters (t+1 to t+4). WACC is as of the end of quarter t. Cash Flow is the sum of next quarter (t+1) or next four-quarter (t+1 to t+4) Operating Income Before Depreciation and Amortization (EBITDA) divided by the average net property plant and equipment for the period. t-statistics in parentheses are computed using two-way clustered standard errors at the firm and year levels.

	Next Quarter Investment (t+1)		**Next Four Quarter Investment (t+1, t+4)**	
WACC	-0.041	-0.028	-0.287	-0.237
	(-2.73)	(-1.98)	(-5.43)	(-4.80)
Cash Flow		0.064		0.114
		(9.26)		(9.81)
Adj. R^2	10.1%	10.4%	29.3%	30.5%
NOBS	215,093	215,093	196,338	196,338

Table 7

WACC vs. Investment: Portfolio Returns

This table presents portfolio results, returns, growth rates, profitability, reinvestment rates and the investment factor (CMA) betas. *Excess Return* refers to average monthly excess return (in excess of Tbill return) in percent, *FF-5 Intercept* is the intercept from Fama and French (2015) 5-factor model consisting of *Market*, *SMB*, *HML*, *RMW* and *CMA* factors. *Investment* is the trailing four quarter investment as defined in Table 6. The 2-way sort based on investment and WACC is an independent sort. The portfolios are constructed each month from January 1989 to December 2022. There are on average about 300 firms in each univariate portfolios and abpout 60 firms in each bivariate portfolio.

Panel A: Univariate Results

	Investment		WACC	
Portfolio	Excess Return	FF 5 Intercept	Excess Return	FF-5 Intercept
Low	1.07	0.03	0.73	-0.07
2	0.91	-0.06	0.84	-0.04
3	0.95	0.00	0.91	-0.02
4	0.82	-0.03	0.99	0.02
High	0.68	-0.11	0.98	-0.04
High - Low	-0.39	-0.14	0.25	0.03
t-stat	-2.77	-1.55	2.14	0.28

Panel B: WACC vs. Investment - Excess Returns

	Low WACC	2	3	4	High WACC	High - Low	t-stat
Low Inv.	0.74	1.07	1.08	1.14	1.24	0.50	3.43
2	0.69	0.85	0.90	1.03	1.01	0.32	1.96
3	0.76	0.82	1.08	1.05	1.02	0.27	1.77
4	0.79	0.77	0.82	0.97	0.78	-0.01	-0.05
High Inv.	0.71	0.70	0.69	0.70	0.62	-0.09	-0.53
High - Low	-0.03	-0.37	-0.39	-0.44	-0.63		
t-stat	-0.16	-2.43	-2.49	-2.23	-3.12		

Panel C: WACC vs. Investment: FF-5 Intercept

	Low WACC	2	3	4	High WACC	High - Low	t-stat
Low Inv.	-0.27	0.09	0.11	0.05	0.15	0.42	3.02
2	-0.13	-0.06	-0.08	0.00	-0.06	0.07	0.42
3	-0.15	-0.07	0.14	0.08	-0.02	0.13	0.79
4	0.13	-0.05	-0.11	0.06	-0.12	-0.25	-1.40
High Inv.	0.04	-0.11	-0.16	-0.11	-0.27	-0.30	-1.68
High - Low	0.31	-0.20	-0.27	-0.16	-0.41		
t-stat	2.04	-1.63	-2.22	-0.99	-2.50		

Panel D: WACC vs. Investment: Cash Flow (EBITDA/NPPE)

	Low WACC	2	3	4	High WACC
Low Inv.	24%	37%	41%	43%	34%
2	34%	51%	61%	61%	49%
3	41%	65%	79%	80%	68%
4	60%	89%	107%	116%	96%
High Inv.	117%	151%	171%	177%	135%

Panel E: WACC vs. Investment: Expected Growth in EBITDA, g_2					
	Low WACC	2	3	4	High WACC
Low Inv.	2%	8%	12%	17%	42%
2	2%	10%	13%	18%	41%
3	3%	12%	15%	20%	45%
4	4%	13%	17%	23%	50%
High Inv.	5%	15%	20%	29%	59%

Panel F: WACC vs. Investment: Reinvestment rate, b					
	Low WACC	2	3	4	High WACC
Low Inv.	0.42	0.28	0.20	0.15	0.13
2	0.65	0.50	0.35	0.27	0.23
3	0.87	0.75	0.55	0.44	0.37
4	0.98	0.94	0.74	0.59	0.48
High Inv.	1.34	1.43	1.26	1.04	0.76

Panel G: WACC vs. Investment: CMA Beta					
	Low WACC	2	3	4	High WACC
Low Inv.	0.20	0.27	0.14	0.15	0.12
2	0.04	0.09	0.07	0.00	0.13
3	0.01	-0.03	-0.02	-0.03	0.10
4	-0.24	-0.11	-0.03	-0.25	-0.04
High Inv.	-0.25	-0.22	-0.32	-0.29	-0.12

Panel H: WACC vs. Investment: RMW Beta					
	Low WACC	2	3	4	High WACC
Low AG	0.06	0.31	0.40	0.38	0.21
2	-0.03	0.37	0.44	0.35	0.15
3	0.13	0.33	0.34	0.23	0.17
4	-0.22	0.22	0.27	0.22	-0.04
High AG	-0.15	0.21	0.21	-0.05	-0.07

Table 8

WACC vs. Asset Growth: Portfolio Returns

This table presents portfolio results, returns, growth rates, profitability, reinvestment rates and the investment factor (CMA) betas. Excess Return refers to average monthly excess return (in excess of Tbill return) in percent, FF-5 Intercept is the intercept from Fama and French (2015) 5-factor model consisting of Market, SMB, HML, RMW and CMA factors. *Asset Growth* is the growth in total assets over the trailing trailing four quarters. The 2-way sort based on Asset Growth and WACC is an independent sort. The portfolios are constructed each month from January 1989 to December 2022. There are on average about 300 firms in each univariate portfolio and about 60 firms in each bivariate portfolio.

Panel A: Univariate Results

	Asset Growth		WACC	
Portfolio	Excess Return	FF 5 Intercept	Excess Return	FF-5 Intercept
Low	1.17	0.14	0.73	-0.07
2	1.01	0.03	0.84	-0.04
3	0.94	0.03	0.91	-0.02
4	0.85	-0.03	0.99	0.02
High	0.49	-0.32	0.98	-0.04
High - Low	-0.68	-0.46	0.25	0.03
t-stat	-4.18	-4.24	2.14	0.28

Panel B: WACC vs. Asset Growth - Excess Returns

	Low WACC	2	3	4	High WACC	High - Low	t-stat
Low AG	0.92	1.10	1.20	1.22	1.29	0.37	2.37
2	0.94	0.84	0.96	1.20	1.09	0.15	1.03
3	0.68	0.88	1.01	1.04	1.05	0.37	2.31
4	0.65	0.83	0.90	1.00	0.97	0.32	1.98
High AG	0.59	0.63	0.53	0.47	0.33	-0.27	-1.50
High - Low	-0.33	-0.47	-0.68	-0.75	-0.96		
t-stat	-1.57	-2.66	-3.68	-3.67	-4.85		

Panel C: WACC vs. Asset Growth: FF-5 Intercept

	Low WACC	2	3	4	High WACC	High - Low	t-stat
Low AG	-0.08	0.16	0.22	0.16	0.19	0.27	1.83
2	0.05	-0.06	-0.01	0.17	0.01	-0.04	-0.26
3	-0.09	0.00	0.09	0.09	0.02	0.11	0.66
4	-0.11	-0.04	-0.05	0.07	0.11	0.23	1.34
High AG	-0.12	-0.18	-0.32	-0.40	-0.55	-0.44	-2.32
High - Low	-0.03	-0.34	-0.54	-0.56	-0.74		
t-stat	-0.20	-2.65	-3.67	-3.49	-4.49		

Panel D: WACC vs. Asset Growth: Cash Flow (EBITDA/NPPE)

	Low WACC	2	3	4	High WACC
Low AG	33%	59%	65%	61%	44%
2	41%	64%	69%	66%	52%
3	48%	71%	79%	80%	64%
4	61%	84%	96%	100%	81%
High AG	70%	101%	111%	113%	88%

Panel E: WACC vs. Asset Growth: Expected Growth in EBITDA, g2					
	Low WACC	2	3	4	High WACC
Low AG	1%	8%	12%	17%	45%
2	2%	9%	12%	17%	42%
3	3%	11%	14%	19%	43%
4	4%	14%	17%	22%	47%
High AG	4%	16%	21%	30%	56%

Panel F: WACC vs. Asset Growth: Reinvestment rate, b					
	Low WACC	2	3	4	High WACC
Low AG	0.46	0.31	0.21	0.14	0.13
2	0.59	0.43	0.33	0.26	0.21
3	0.90	0.75	0.57	0.44	0.38
4	1.19	1.08	0.82	0.63	0.53
High AG	1.30	1.40	1.24	1.01	0.73

Panel G: WACC vs. Asset Growth: CMA Beta					
	Low WACC	2	3	4	High WACC
Low AG	0.27	0.25	0.20	0.13	0.29
2	0.08	0.22	0.18	0.13	0.16
3	-0.12	0.05	-0.02	-0.09	0.01
4	-0.17	-0.10	-0.10	-0.23	-0.16
High AG	-0.34	-0.39	-0.44	-0.37	-0.24

Panel H: WACC vs. Asset Growth: RMW Beta					
	Low WACC	2	3	4	High WACC
Low AG	-0.05	0.11	0.26	0.31	0.10
2	0.06	0.34	0.39	0.33	0.30
3	0.05	0.37	0.37	0.28	0.22
4	0.08	0.34	0.35	0.25	-0.01
High AG	-0.17	0.19	0.21	0.00	-0.15

Table 9
Regression of Aggregate Investment on Aggregate WACC

This table presents time-series regressions of future aggregate investment on aggregate WACC using quarterly data from 1989-Q1 to 2022-Q4. Panel A presents results for the next quarter's investment and Panel B presents results for the next four quarter investment, where investment is defined in Table 6. The aggregate WACC is the enterprise value-weighted firm level WACC at the end of quarter t. d is a pandemic dummy variable with a value of 1 for the period 2020-Q1 to 2022-Q4 and 0 for all other periods. *Term Spread* is the yield on AAA rated bonds minus the yield on 3 month T-bill rate. *Default Spread* is the yield on BAA rated bonds minus the yield on AAA rated bonds. The yields are obtained from FRED (Federal Reserve of St. Louis research database). *Market Return* is the lagged quarterly value-weighted market returns from CRSP. t-statistics in parenthese are computed using Newey-West standard errors with 12 lags.

Panel A: Next Quarter Investment (t+1)			
WACC	-0.263	-0.266	-0.077
	(-2.96)	(-3.20)	(-1.28)
d	-0.017	-0.021	-0.019
	(-4.53)	(-5.92)	(-6.72)
Cash Flow		0.355	0.317
		(7.18)	(5.81)
Term Spread			-0.268
			(-2.74)
Default Spread			-0.444
			(-1.70)
Market Return			-0.027
			(-3.49)
Adj.R^2	14.3%	35.7%	44.4%
NOBS	136	136	136
Panel B: Next Four Quarter Investment (t+1, t+4)			
WACC	-0.591	-0.934	-0.157
	(-1.59)	(-2.27)	(-0.57)
d	-0.044	-0.081	-0.068
	(-3.12)	(-5.12)	(-5.88)
Cash Flow		0.433	0.390
		(8.83)	(6.95)
Term Spread			-0.957
			(-2.32)
Default Spread			-2.358
			(-2.54)
Market Return			-0.034
			(-1.70)
Adj.R^2	7.4%	44.3%	58.1%
NOBS	133	133	133

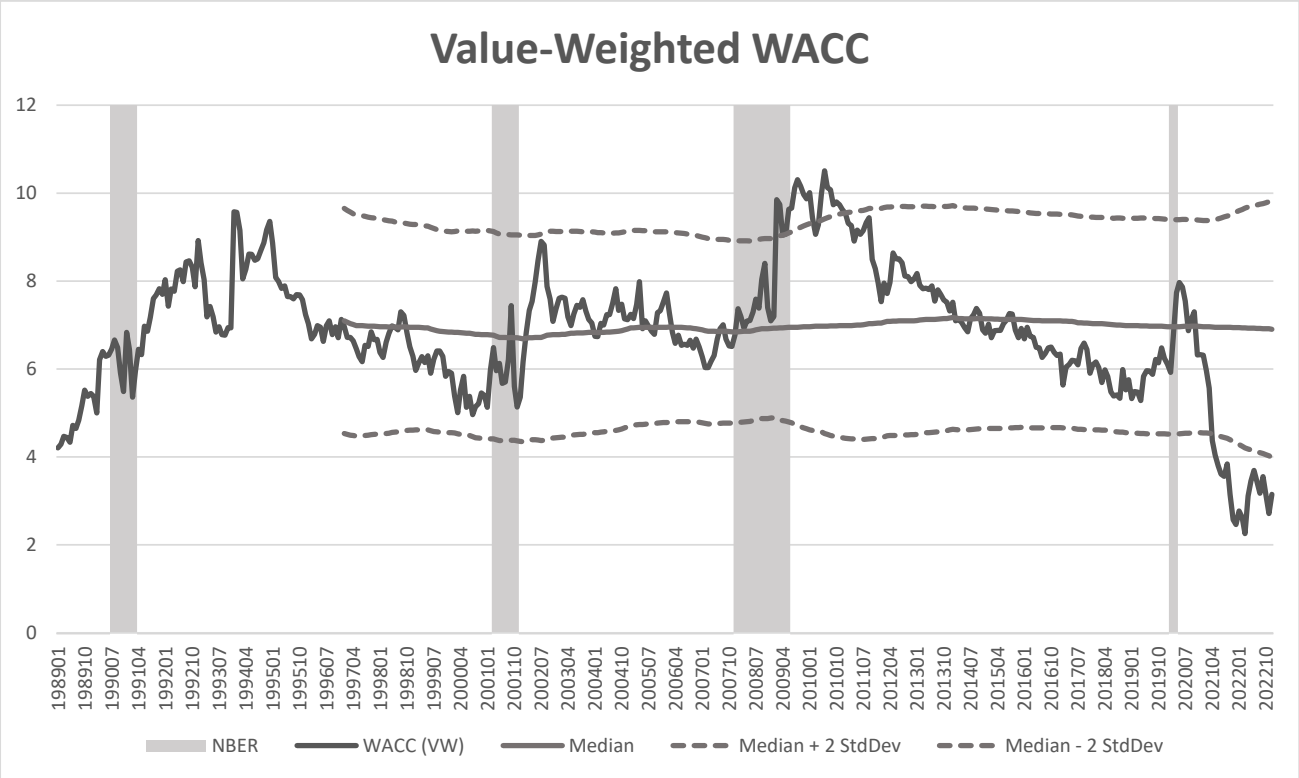

Figure 1a: Value-Weighted Real WACC from January 1989 to December 2022

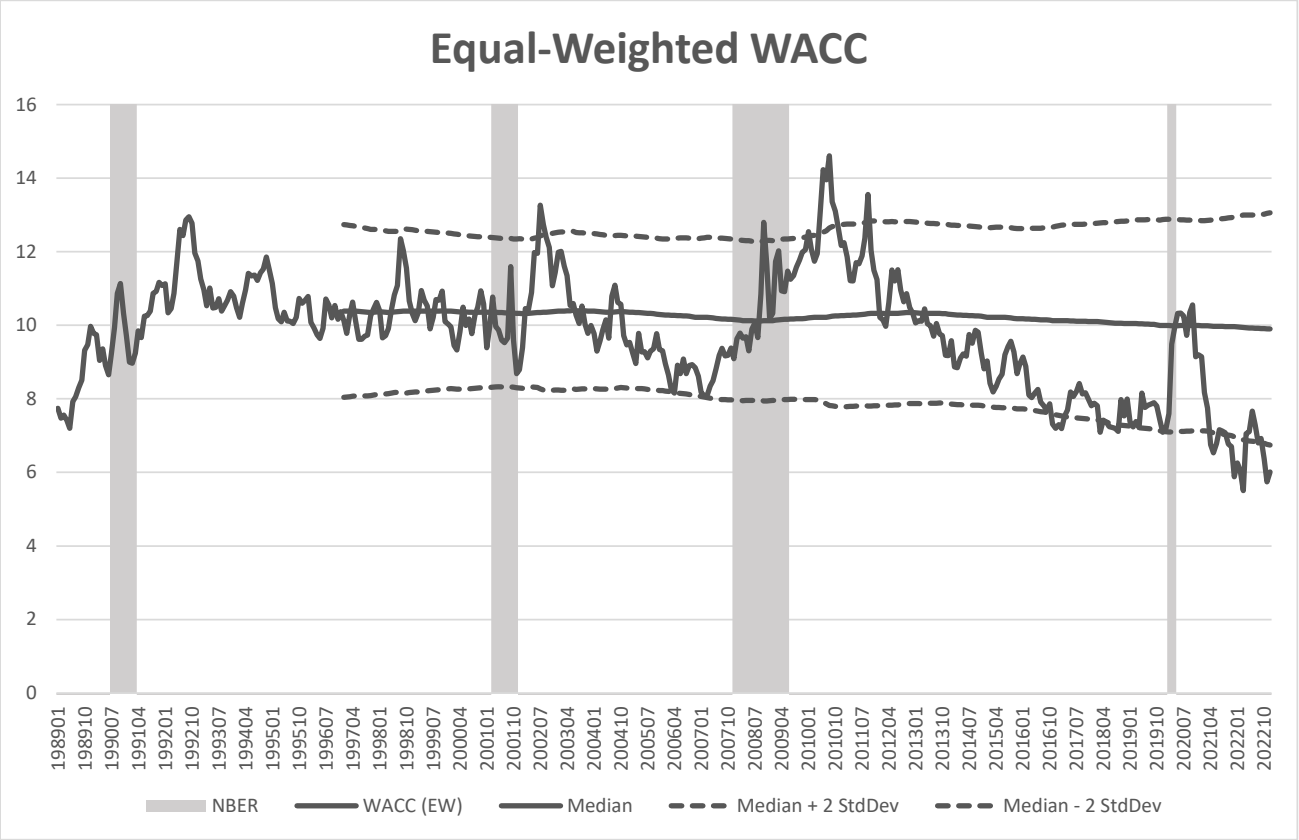

Figure 1b: Equal-Weighted Real WACC from January 1989 to December 2022

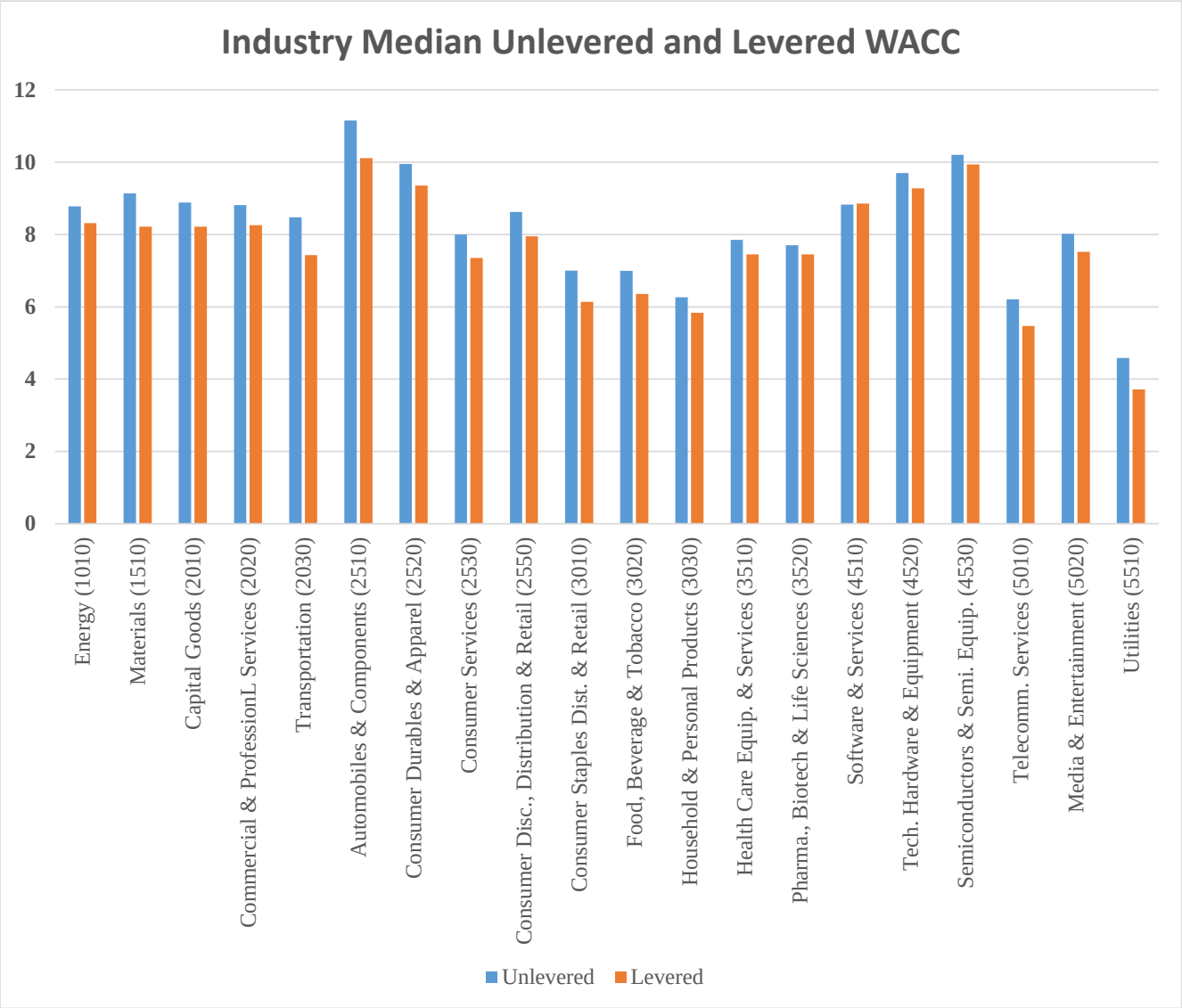

Figure 2 displays the median levered and unlevered WACC for various GICS 4-digit industries. The levered WACC is the estimated real WACC. The unlevered WACC is based on Modigliani and Miller formula relating unlevered WACC (cost fo equity) and levered WACC: Unlevered WACC = Levered WACC/(1 - Tax Rate * Leverage Ratio) where leverage ratio is the *market leverage* defined in Table 4.

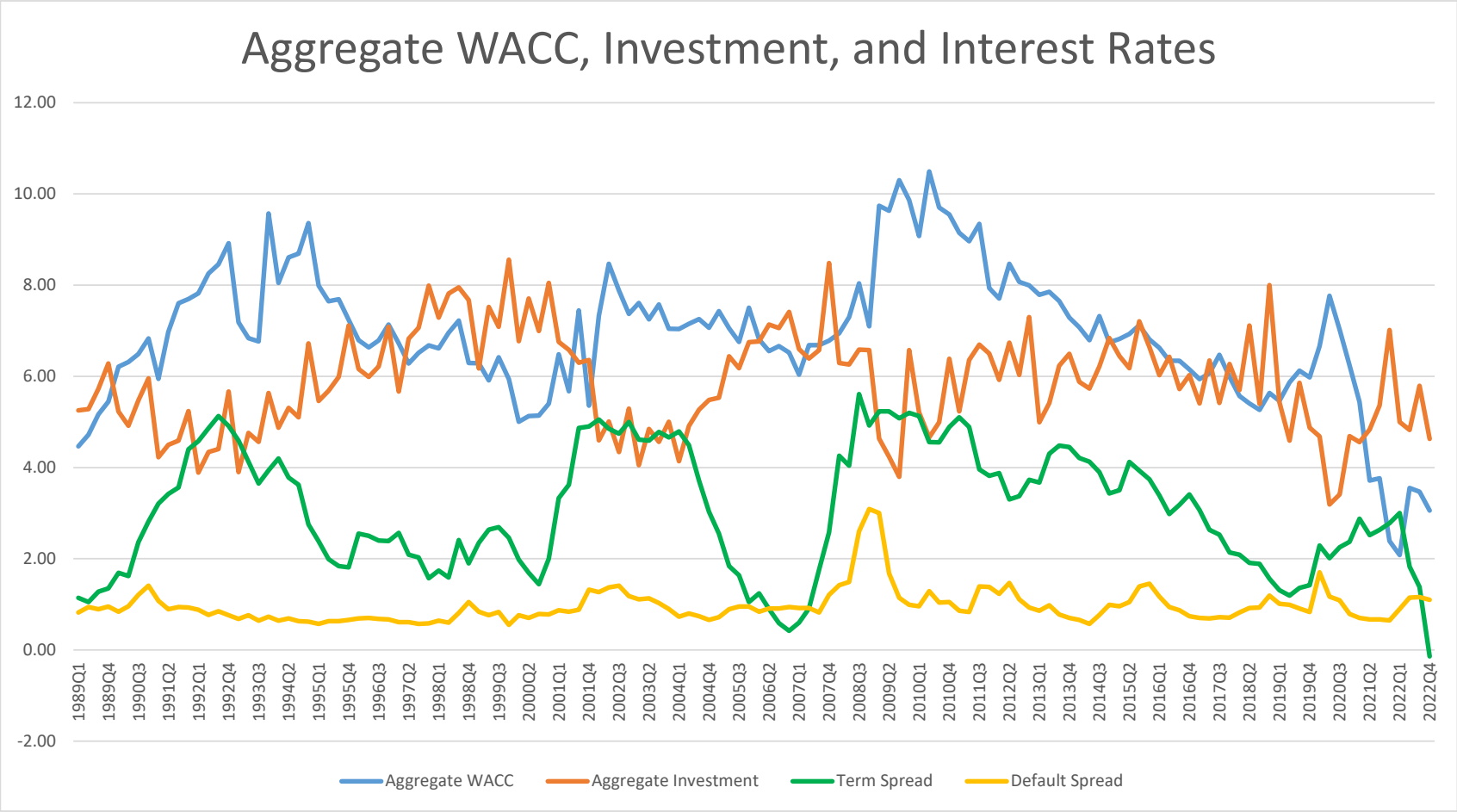

Figure 3 plots the aggregate investment (%) and aggregate real WACC (%) from 1989-Q1 to 2022-Q4. The aggregate WACC is the enterprise value weighted average of the individual WACCs at the end of each quarter. Aggregate investment is the sum of firm-level investment (CapEx + Cash Acquisitions - Net sale of property) for each non-financial, non-utility firm with common stock listed on the NYSE/AMEX/Nasdaq, divided by the average net property, plant and equipment dueign the quarter. Term Spread is defined as AAA Yield - 3 month T-bill Yield. Default Spread is defined as BAA Yield - AAA Yield.